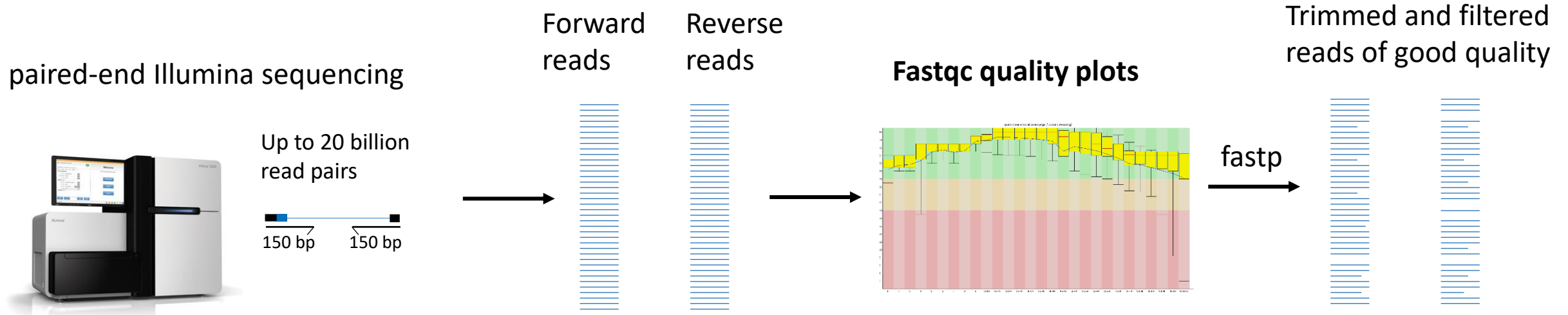


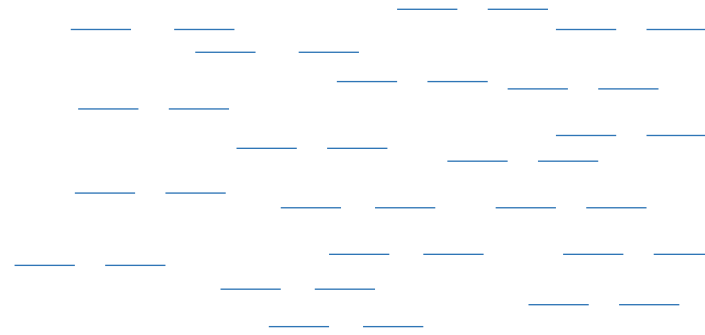
Mapping to a
reference genome

Yesterday: Quality check and trimming raw reads

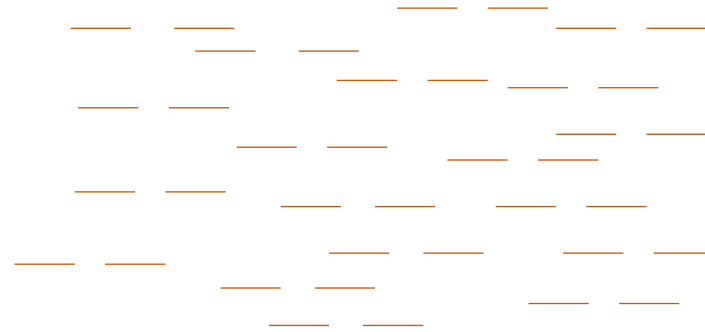


Today: Alignment to the reference genome

reference



bam individual 1



bam individual 2

Aligning/Mapping short read data to a reference genome



bwa-mem2 (Burroughs-Wheeler Alignment with maximum exact matches)

Version 2 is 50% faster than bwa-mem

Very new tool worth considering: minibwa
(<https://arxiv.org/abs/2606.15357>) which is much faster, similar quality and even better at aligning reads with indels

Regions that are repeated in the genome make it very difficult to map reads

Reference genome



Read to be aligned to the reference



Regions that are repeated in the genome make it very difficult to map reads

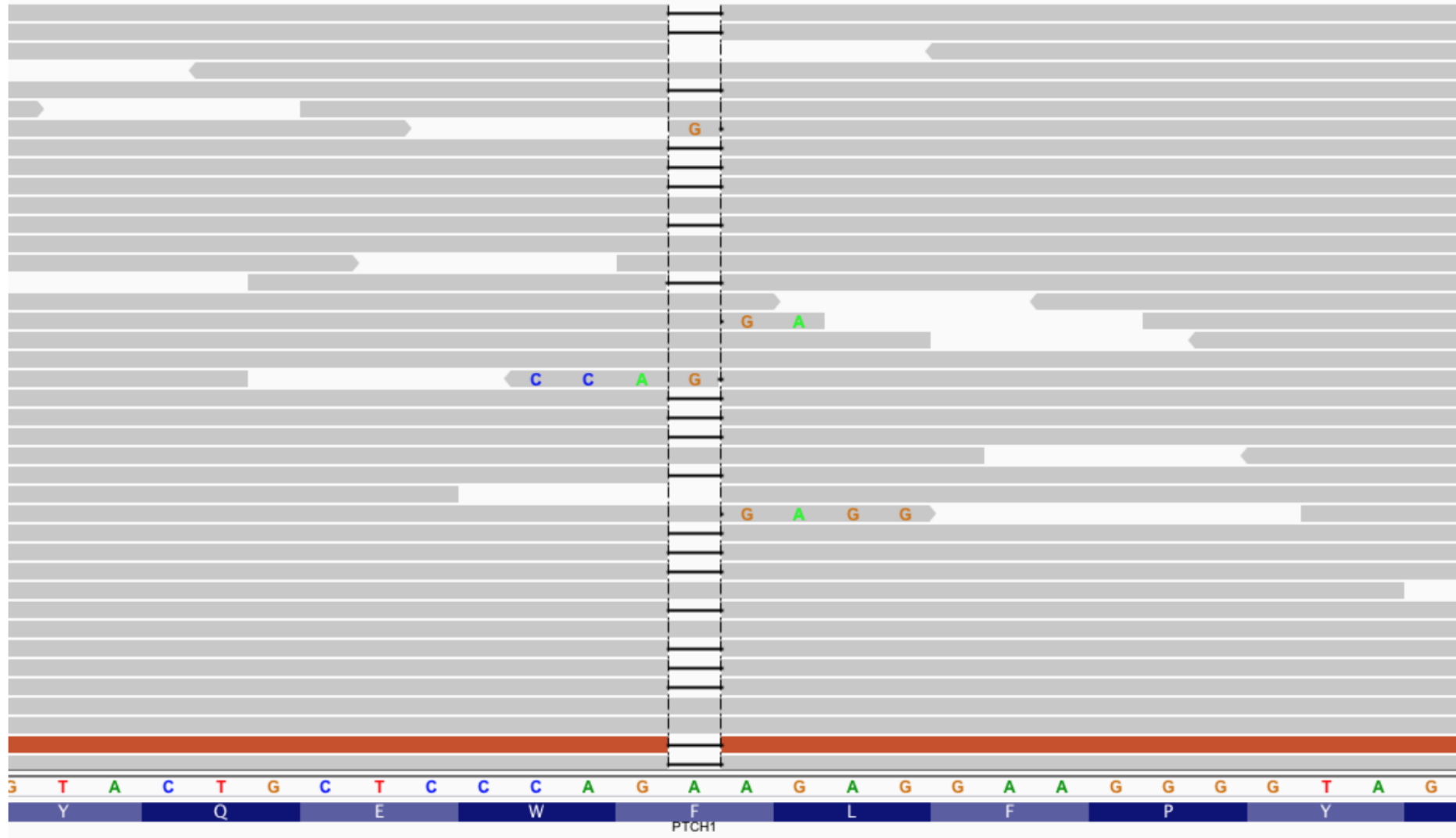
Reference genome



Read to be aligned to the reference

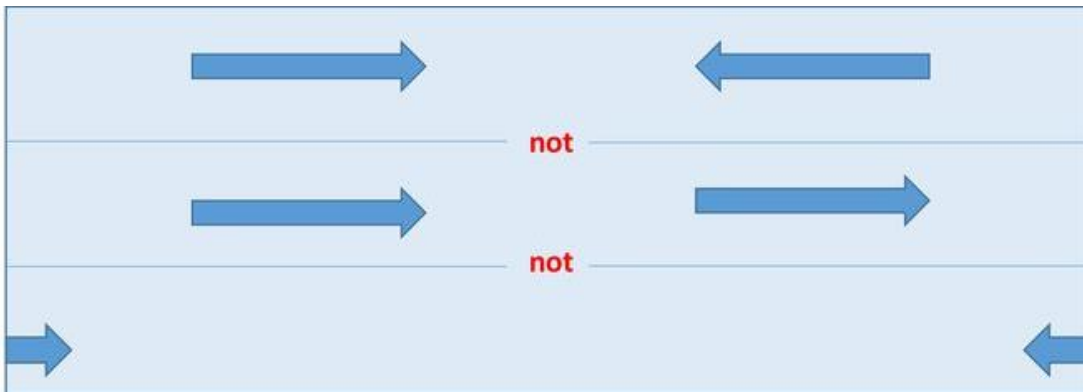


Indels: deletions and insertions



Aligning reads is complicated by:

- Imperfect match to the reference due to mutations or sequencing errors, or errors in the reference genome
- Multiple positions where the read could match (repeated regions)
- Low quality of the read
- With paired-end reads: only one read maps or the other read maps on a different chromosome



Alignment tools such as bwa-mem2 are able to handle all of these complications and give information on the mapping quality.

General principle

- The reference genome is indexed to make mapping faster
 - Reference: ATCATTCATCTG (would of course be much longer)
 - Index: for each k-mer figure out where they are in the genome
e.g. for k=3, k-mers are combinations of 3 nucleotides

Positions of 3-mers in our toy example:

ATC: 1,8 (ATCATTCATCTG)

TCA: 2 (ATTCATTCATCTG)

CAT: 3,7 (ATCATTCATCTG)

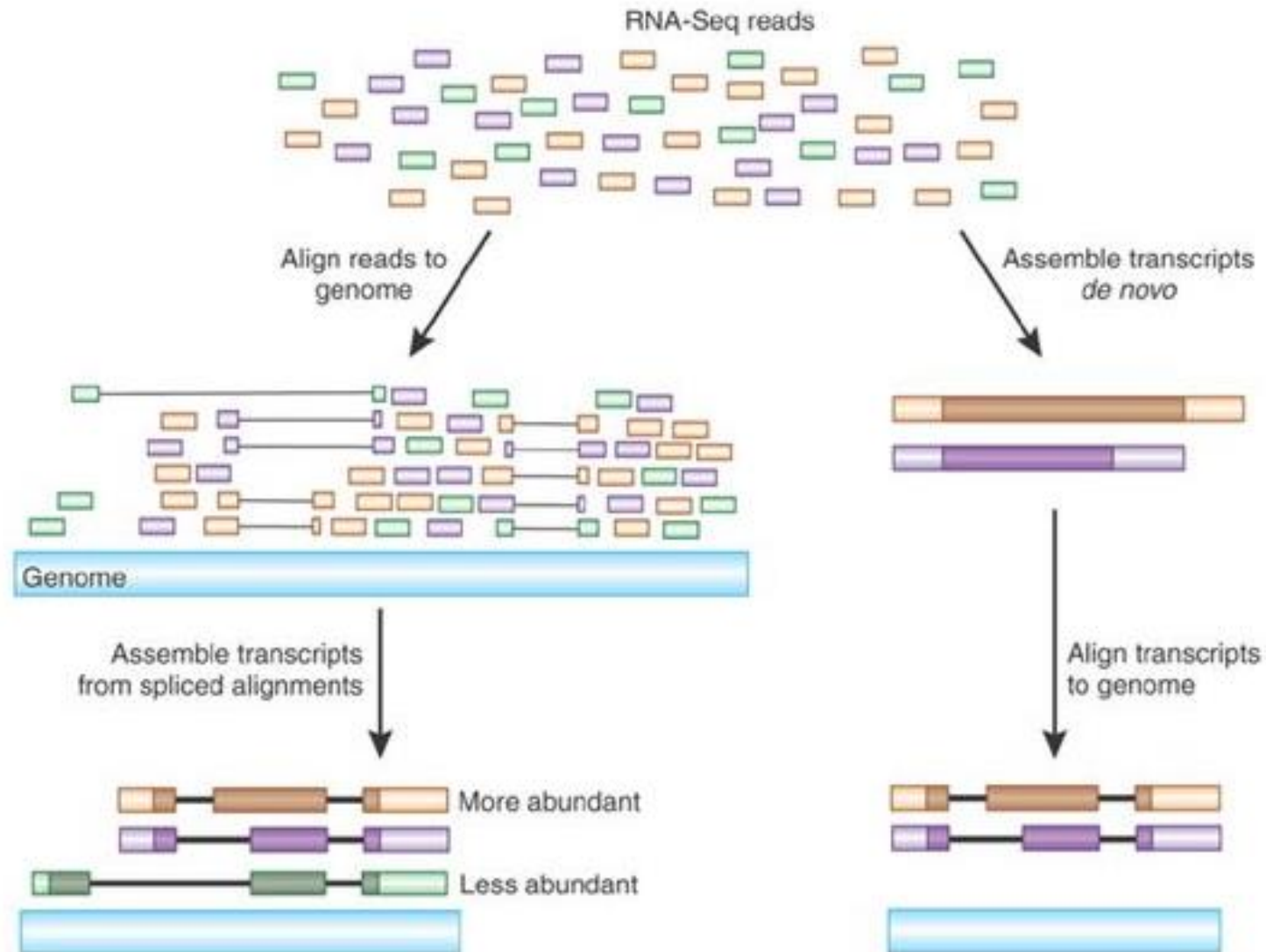
ATT: 4 (ATCATTCATCTG)

etc.

- The reads are mapped using the index:
 - Read example: CATTC
 - First k-mer (CAT) suggests the read could start at position 3 or 7
 - Second k-mer shows that it matches (ATT) at position 4

Aligning other type of data: RNA data

Tools:
e.g.
STAR
HISAT2



Checking if a genome exists: <https://goat.genomehubs.org>

Genomes on a Tree (GoaT)

Genomes on a Tree (GoaT): A versatile, scalable search engine for genomic and sequencing project metadata across the eukaryotic tree of life. Challis *et al.* 2023. Wellcome Open Res 2023, 8:24 [doi:10.12688/wellcomeopenres.18658.1](https://doi.org/10.12688/wellcomeopenres.18658.1)

GoaT has been built using [GenomeHubs](#) to help coordinate efforts across the [Earth Biogenome Project](#) (EBP) Network at all stages from planning through sequencing and assembly to publication. [read more...](#)

Search GoaT

Type to search GoaT taxon index (e.g. Canidae)

 TAXON



include descendants



include estimates



empty columns



result columns



query builder



clear all



[Try the new search box](#)

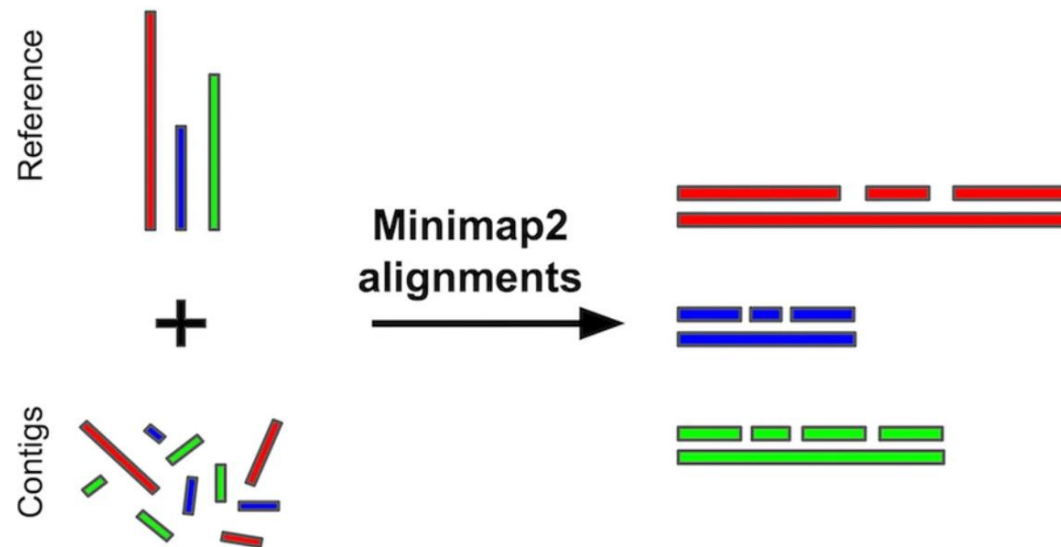
[show examples](#) or [browse tree](#)

Improving a draft genome assembly with a chromosome-level reference genome of a close relative

Reference-guided scaffolding with RagTag (successor of RAGOO)

<https://github.com/malonge/RagTag>

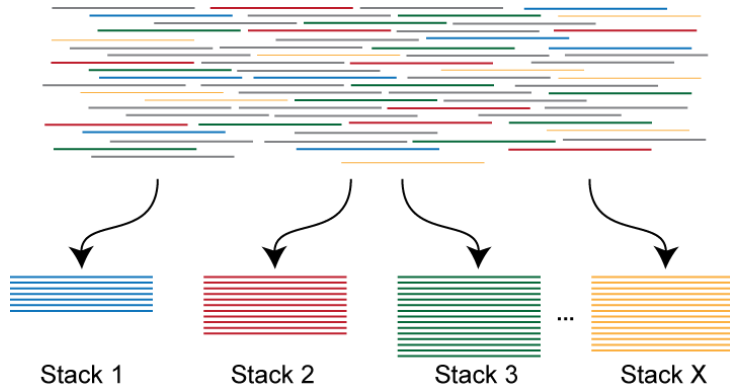
Uses minimap2 for genome alignment



RAD data without a reference genome

Make your own reference with:

- **STACKS**: for datasets of <https://catchenlab.life.illinois.edu/stacks>




Stacks

- **iPyRAD** (for RAD, ddRAD, GBS, 2bRAD): for datasets with different species <https://ipyrad.readthedocs.io/en/master/index.html>

PacBio mapping

Karin Näsvall



Sequel IIe Instrument

Generate Sequencing Data

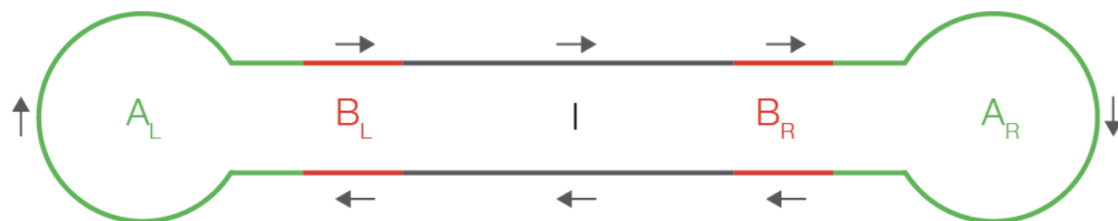


Data Transfer/SMRT Link Server



SMRT Link GUI

Access using a web browser



A schematic drawing of a SMRTbell

Automatic HiFi reads generation (Export Reads)

`hifi_reads.fastq.gz` - FASTQ file containing HiFi Reads

`hifi_reads.fasta.gz` - FASTA file containing HiFi Reads

`hifi_reads.bam` - BAM file containing HiFi Reads

HiFi reads QV > 20

A =Adaptors, B=Barcode, I=insert (DNA of interest)

<https://pacbiofileformats.readthedocs.io/en/13.0/>

PacBio

```
cat m84093_240426_124306_s1.ccs.bc2033.stats
```

```
-
```

```
A = 7108650862 (34.9%), C = 3058767823 (15.0%), G = 3061827434 (15.0%), T = 7127816130 (35.0%),  
CpG = 1003156910 (4.9%)
```

```
sum = 20357062249, n = 2218131, mean = 9177.57438537219, largest = 30301, smallest = 135
```

sum = total number of sequences

n = number of sequences

mean, largest, smallest = length of reads

PacBio

<https://pacbiofileformats.readthedocs.io/en/13.0/BAM.html#hifi-reads>

```
$ samtools view /pacbio/m84093_240426_124306_s1.ccs.bc2033.bam | head -n2
```

m84093 240426 124306 s1/261297081/ccs 4 * 0 255 * * 0 0
GTTTGGTTATATTGTTTCATAGACACCTACGGTACACGAATATAATAAAACAAATAATATTAATTGTTATACATATTTAAATATAAAATTAATATTGTAGCAACTATTAACATACTAGTTCAC
TTCCTTCAACAGTTTAAATCAAGGATTCCCTTAACCTATTTTTTACCAAAACGTAAGTATGATACCAAGATTGTTTTCTAATGATTTTGATGATATGGATGTGATTTGACTACATTAATGCTA
CTGCTAGAACTTTACGATATCCCTCACATCACCTCGTCAAATGCTTTGTTTAAATTGTTAAAAAAATTCTTTGTATTTAAAGTGAGTAATCCTTAAGTAAACAGTGTTTCAAAGT
TAAAAATTTATTTTATACGTGAAATGTCGATATTATCGATGGCATTATTAGTTTCCAGCATATTTATAAAAGCCTTTGTGACGCAATCACCGAGCGGTGAAAATATGATTTGAGACGCT
TCCTGAAATTGAAAAAAGAAACCCCTCTATTGTTTTTAGATGATGTTTATTTTCTGCATTACTTTCTTAAATTAATATAATTTATACTATACAAATGTGACATTTCAATTGCGATACGA
GTTTAAAGAGGAGAACATCGATTACTGAAGTTATACAACTTCTCCACAATAGATAACATTAACCTTTATATTTTAAACTTTAAGATAGTTCTTAAAGTGTTGAAAAGATGTAAAAAAT
GGTACATTTGAAGAAACCTTTATTTATGAAGAGTTTAAAGGTTATCCGTTATTTTCAAAAAGATTTTACATCTTTCTAATGCGAAATGTTAAAAATCTTAAAAATTTTTTAGCTTAT
TTAACGAATAAGAAGAGAAAATATTTATTAATAATTCATATTTGATATTTTATTTATTGATTTTATTTATTCTTTTAAATAGTTTAAATTTAGTTTCGATTTGAAATCTCTTTTATTGT
TAAATGTTTCATCTGGAGACATTTACAGTCCAATAGTTTCCAAAAGAAATTTTAAACGCTGTTTCGAAATTATCTTTCGTGATTATGAAGTGTAAATTTCCACTCAGTTAATG
GTTTAAATTTTATGTATTGCAAGAAAAGATAATAAAAGCTTGAAATAAAATCATTTTATCAGTATTTTGCATACCTATAAATATTTGAAGGTA

— — — —

$$\begin{aligned} & \left(\sum_{i=1}^n |D_i| + \sum_{i=1}^n |D_i| + 7 \sum_{i=1}^n |D_i| + \sum_{i=1}^n |D_i| + 2 \sum_{i=1}^n |D_i| + 7 \sum_{i=1}^n |D_i| \right) \\ & \left(\sum_{i=1}^n |D_i| + \sum_{i=1}^n |D_i| + 7 \sum_{i=1}^n |D_i| + \sum_{i=1}^n |D_i| + 2 \sum_{i=1}^n |D_i| + 7 \sum_{i=1}^n |D_i| \right) \end{aligned}$$

ac:B:i,8,0,10,0 bl:Z:CGACTCGTATGAGTAI bt:Z:ATACTCATACGAGTCG bx:B:i,16,16 ec:f:7.5012 ma:i:0 np:i:7 ql:Z:||||| qt:Z:||||| rq:f:0.999815
sn:B:f,17.2518,25.3332,6.55462,11.2707 we:i:8638323 ws:i:40959 zm:i:261297081 qs:i:26 ge:i:17132 bc:B:S,0,0 bq:i:100 cx:i:12
ls:B:C,134,162,98,99,164,48,45,45,48,162,98,113,100,164,108,101,97,100,130,162,113,108,186,73,73,73,73,73,73,73,73,73,73,73,73,
73,73,73,73,73,73,162,115,113,186,65,65,71,67,65,71,84,71,71,84,65,84,67,65,65,67,71,67,65,71,65,71,84,65,67,84,166,110,101,115,116,101,100
,192,164,112,53,109,99,0,165,116,114,97,105,108,130,162,113,108,186,73,
,162,115,113,186,65,71,84,65,67,84,67,84,71,67,71,84,84,71,65,84,65,67,67,65,67,84,71,67,84,84 RG:Z:f56a67e5/0--0

PacBio mapping – Minimap2

Minimap2 <https://github.com/lh3/minimap2>

Uses **minimizers** – short sequences of length $-k$ [default 15 bases]

1. Extracts minimizers from the reference (target) and index them
2. Match each minimizer in the query sequence against the reference set of minimizers
3. Sorts the minimizers after genomic position
4. Make a chain of overlapping minimizers
5. Repeat for all query sequences

```
Minimap2 -ax map-hifi target.fa query.fa > output.sam
```

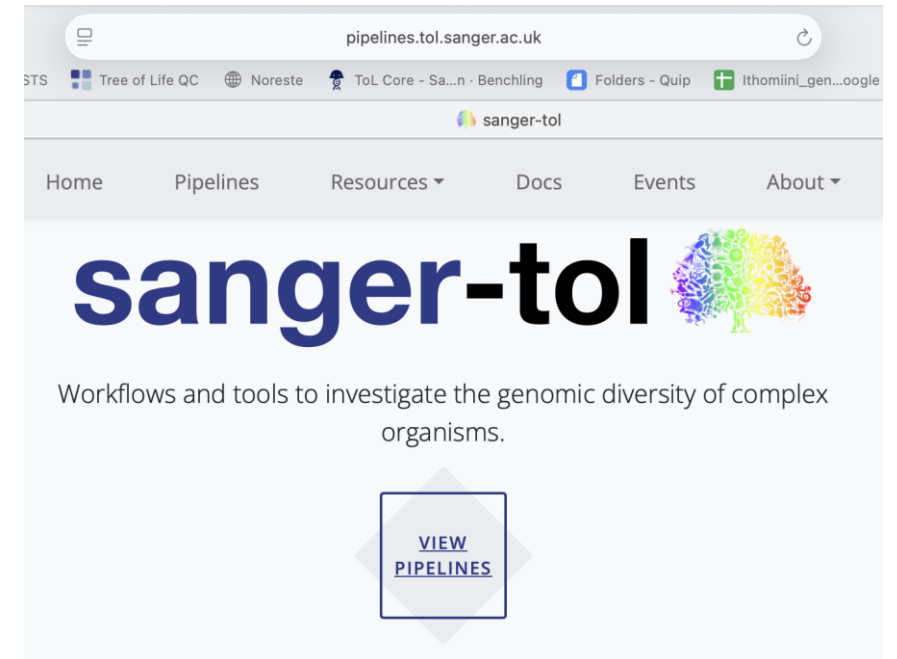
```
Align PacBio high-fidelity (HiFi) reads to a reference genome (-k19 -w19 -U50,500 -g10k -A1 -B4 -O6,26 -E2,1 -s200)
```

Readmapping pipeline

Preprocessing - filtering
and adaptor removal

Alignment - minimap2

Alignment postprocessing
- mapping statistics



<https://pipelines.tol.sanger.ac.uk>